

Tax Cuts and Innovation: Unpacking Ireland's Finance Act 2015

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Abstract

In 2016, Ireland introduced the *Knowledge Development Box* (KDB) through the *Finance Act 2015*, reducing the tax rate on income derived from intellectual property (IP) from 12.5% to 6.25% to foster innovation. This study examines the impact of the KDB on the number of patents filed and on the growth rate of the depreciation of IP assets by Irish firms. Using advanced econometric methodologies, including static and dynamic Difference-in-Differences (DiD) models as well as the Synthetic Control Method (SCM), we assessed the effect of this tax reform. Our analyses reveal a significant decrease in the number of patents filed and an increase in the depreciation rate of IP assets following the implementation of the KDB. These results suggest that the tax reform may have negatively influenced innovation behaviors and altered the management of IP assets by Irish firms. However, the violation of the parallel trends assumption and the inherent limitations of the available data qualify the robustness of these conclusions, making it difficult to establish a direct causal link between the KDB and the observed changes. This study highlights the complexity of interactions between tax policies and innovation. While the reform appears to have impacted the number of patents filed and the management of IP assets, it is essential to consider the methodological challenges and data limitations in future studies to further explore these relationships and better understand their implications.

Keywords: Knowledge Development Box, intellectual property, tax reform, innovation, research and development, difference-in-differences, synthetic control method, asset depreciation, patents, Ireland.

1 Introduction

In the era of globalization and rapid technological innovation, nations engage in intense competition to attract and retain research and development (R&D)-intensive companies. Intellectual property (IP), encompassing patents, trademarks, copyrights, and other intellectual creations, has become an essential strategic lever to boost economic competitiveness and foster sustainable growth. Tax policies thus emerge as crucial tools to encourage investment in IP and position a country as an innovation hub. In 2015, Ireland introduced the *Knowledge Development Box* (KDB)¹, an innovative tax regime designed to encourage companies to develop and exploit their IP assets within Irish territory. This reform, formalized by the *Finance Act 2015*², reduces the tax rate on qualified IP-derived income to 6.25%, in accordance with the standards of the Organisation for Economic Co-operation and Development (OECD). Prior to the introduction of the KDB, the standard rate of 12.5% was already competitive but offered limited incentives to stimulate intensive investments in R&D. However, research on the effectiveness of such tax reforms, particularly their ability to stimulate innovation, has sparked significant debates among economists and policymakers. In this context, this study contributes to shedding light on these issues by examining the impact of the KDB in Ireland on innovation and the management of intangible assets.

Despite its ambition, the effectiveness of this type of tax reform in promoting innovation remains a subject of debate, particularly concerning their real impact on inventive activity and the management of intangible assets. This study seeks to answer a key question: to what extent has the KDB influenced innovation in Ireland, evaluated through two main indicators—the number of patents filed and the valuation of IP assets in companies' balance sheets? To address this issue, we formulate two main hypotheses: (1) the KDB should have led to an increase in the number of patents filed, reflecting intensified R&D activities; and (2) it should also have induced a rise in the depreciation rate of IP assets, indicating an increased valuation of these assets in companies' balance sheets. To test these hypotheses, we employed rigorous econometric methodologies, including static and dynamic difference-in-differences (DiD) models (event study)³, as well as the synthetic control method (SCM)⁴. These approaches not only allow us to control for fixed effects and temporal trends but also to examine the validity of the parallel trends assumption, fundamental for estimating a causal effect in DiD models⁵. We first verified the linear trends⁶ to assess compliance with this assumption, then applied the econometric models to estimate the effect of the reform.

The results of our analysis reveal unexpected findings. Contrary to our initial hypothesis, the static DiD model indicates a significant decrease in the number of patents filed after the introduction of the KDB. This decrease is also confirmed by the event study and the SCM, although the violation of the parallel trends assumption limits the scope of causal conclusions. Regarding the growth rate of IP asset depreciation, the results suggest an increase post-reform, which may indicate an increased valuation of these assets by companies. These observations raise important questions about the effectiveness of tax incentives in innovation and the strategic behaviors of companies in response to such reforms. They highlight the complexity of the underlying mechanisms of innovation and the need for in-depth analysis to fully understand the impact of tax policies on economic activity. In this perspective, our study makes a significant contribution by exploring not only the immediate effects of the KDB on innovation in Ireland but also by discussing the broader implications for public policies and future directions in taxation and innovation.

The remainder of this article is organized as follows: Section 2 provides a review of the relevant literature, establishing the theoretical and empirical framework of this study. Section 3 describes the data used and the methodology adopted, while Section 4 details the methodology (econometric strategy), including the DiD models, dynamic DiD, and SCM, and examines the key assumptions, notably the parallel trends assumption with the linear trends model. Section 5 presents the results of the empirical analysis, highlighting the main estimated effects. Finally, Section 6 discusses the conclusions, their policy implications, and proposes avenues for future research.

To place this study in its academic context and explore the theoretical and empirical foundations of the impact of tax incentives on innovation, it is essential to examine the major contributions of the economic literature on the subject.

2 Literature

The relationship between tax policies and innovation has been extensively studied in the economic literature. Tax incentives are considered potential tools to stimulate investment in R&D and, consequently, to increase innovation activity within an economy (Hall et al., 2019; Bloom et al., 2002)^{7,8}. In this context, tax regimes specific to IP income, such as *patent boxes*, have garnered growing interest among researchers and policymakers. This study fits into this debate by specifically analyzing the impact of the Irish KDB on innovation, within the broader framework of tax policies aimed at stimulating R&D activities. Tax incentives for R&D are designed to reduce the marginal cost of investment in innovation, thereby encouraging companies to increase their R&D expenditures (David et al., 2000)⁹. Empirical studies have shown that these incentives can have a significant positive effect on innovation activity (Bloom et al., 2002; Ernst et al., 2014)^{8,10}. For example, Bloom et al.⁸ found that R&D tax credits substantially increase companies' R&D expenditures, which translates into an increase in the number of patents filed. However, the effectiveness of tax incentives depends on their design and implementation. Tax regimes focused on IP income, such as *patent boxes*, have been criticized for potentially favoring profit localization rather than genuine innovation activity (Griffith et al., 2014)¹¹. Griffith et al.¹¹ pointed out that these regimes can encourage companies to transfer their IP rights to low-tax jurisdictions without necessarily increasing R&D expenditures or patenting activity. *Patent boxes* are tax devices that offer reduced tax rates on income derived from IP (Evers et al., 2015)¹². Their objective is to incentivize companies to locate their IP assets and associated income in the country offering the preferential regime. The introduction of such regimes in several European countries has led to a series of studies evaluating their impact on innovation. Ernst and Spengel¹³ analyzed the effect of *patent boxes* on patenting activity and found mixed evidence regarding their effectiveness in stimulating innovation. They noted that while these regimes can attract IP rights, they do not necessarily lead to an increase in local R&D activities. Moreover, Alstadsaeter et al.¹⁴ highlighted that *patent boxes* can lead to harmful tax competition between countries, without clear benefits in terms of increased innovation. The use of robust econometric methodologies is essential to evaluate the impact of tax policies on innovation. DiD models are commonly used to estimate causal effects by comparing changes in outcomes between a treated group and a control group before and after an intervention (Angrist et al., 2008)⁵. However, the validity of this approach relies on the parallel trends assumption, which can be violated in certain contexts. The SCM offers an alternative by constructing a synthetic counterfactual that reproduces the characteristics of the treated group before the intervention (Abadie et al., 2010)⁴. This method has been successfully applied to evaluate the impact of various economic and fiscal policies (Abadie et al., 2015)¹⁵. Despite the abundance of research on *patent boxes* and tax incentives for innovation, there is a need for specific studies on the impact of the Irish KDB, particularly regarding its effect on the number of patents and the management of IP assets. Most existing studies focus on the overall effects of tax incentives without deeply analyzing the specific mechanisms by which these policies influence companies' innovation behavior. Our study contributes to the literature by providing an original and detailed empirical analysis of the impact of the KDB in Ireland on two key indicators: the number of patents filed and the growth rate of IP asset depreciation. By using advanced econometric methodologies, we bring new perspectives on the effectiveness of this type of tax regime and how it influences companies' strategic decisions regarding innovation and the management of intangible assets. Given the challenges associated with the potential violation of the parallel trends assumption in DiD models, the combined use of dynamic DiD and SCM strengthens the robustness of our analysis. The dynamic DiD allows capturing the delayed temporal effects of the reform, while the SCM offers an alternative to construct a more precise counterfactual when the conditions of the classic DiD are not satisfied^[4,2,3].

In summary, the existing literature underscores the importance of tax policies in stimulating innovation, while cautioning against the limitations and potential adverse effects of regimes such as *patent boxes*. Our research aims to fill a specific gap

by examining the impact of the KDB in Ireland, providing an original contribution to the understanding of the effects of these policies on patenting activity and the management of IP assets.

3 Data

This section presents the collection, selection, and preparation of the data used to evaluate the impact of the introduction of the KDB in Ireland on the number of patents filed and the depreciation rate of IP assets of Irish companies. We will describe the main data sources, the criteria for selecting companies and control countries, as well as the key variables included in our analysis. Additionally, we will discuss the methods employed to ensure data quality and reliability, thus laying the groundwork for the methodological approaches adopted, namely the DiD, dynamic DiD (event study), and the SCM.

Data selection

The data were collected from the websites of the Organisation for Economic Co-operation and Development (OECD)¹⁶ and the World Bank¹⁷. These sources are recognized for their reliability in providing economic, energy, and environmental statistics.

Selection of control country for patents^[Appendix8.1]

In the context of the DiD analysis, the choice of control country was made by considering structural similarities with the treated country as well as the temporal proximity of the analyzed reform. To do this, we evaluated several candidates by visually ^[Appendix8.1] comparing the trends of key variables and examining the discrepancies with Ireland.

Among the countries considered, Switzerland was chosen as the control country for the patent analysis due to its structural and temporal proximity to Ireland in this domain. For the analysis of the depreciation of IP assets, Finland was selected because of its suitability in capturing comparable dynamics. Other initially evaluated countries included Sweden and Austria, each analyzed according to similar criteria. The selection of these countries reflects their relevance in enhancing the robustness of the analyses conducted.

Observation period

The exclusion of data after 2018 is based on the fact that the subsequent period was marked by major disruptions related to the COVID-19 pandemic, which profoundly altered economic and social dynamics. These changes, such as lockdowns and the drastic reduction of industrial activities, could have significantly affected patent filings and the depreciation of IP assets. Including these data could introduce significant biases, thus distorting the analysis of the impact of the KDB reform for the studied period from 1996 to 2018.

3.1 Key variables for patents

The variables selected for the analysis aim to evaluate the impact of the KDB tax reform in Ireland, particularly on patent dynamics. These variables, chosen for their economic and technological relevance, allow for a scientific exploration of the links between tax incentives, innovation, and national competitiveness. They include:

Patents¹⁸ in number, (OECD): A measure of technological innovation and inventive activity, a key indicator of economic competitiveness. *Data transformation: in logarithm*

Net Trade Balance (NX)¹⁹ in billions of US dollars, Purchasing Power Parity (PPP) converted base 2022, (OECD): An economic indicator reflecting the balance between exports and imports, linked to overall economic performance. *Data transformation: in growth rate (%)*

Real GDP²⁰ in acceleration (second derivative, % growth rate of growth rate), (OECD): A measure of the change in the economic growth dynamics, reflecting shifts in economic momentum and trends.

Population²¹ in millions, (OECD): An essential demographic indicator for assessing the scale of economic and social activities. *Data transformation: in logarithm*

R&D Expenditure²² as % of GDP, (OECD): A measure of the national effort in R&D, an indicator of competitiveness and innovation. *Data transformation: in growth rate (%)*

In addition to determining an adequate observation period, it is essential to choose an appropriate control country for our DiD analysis to ensure the robustness and validity of the results obtained.

3.1.1 Descriptive statistics

The descriptive statistics of the key variables for Ireland and Switzerland are presented below. These statistics include the means, medians, standard deviations, skewness, kurtosis, and the 25% and 75% quantiles.

	mean	median	std	skewness	kurtosis	25%	75%
patents	5.899679	5.620342	0.382714	-2.458030	1.311031	5.620342	6.197399
gdp	0.021203	0.021203	2.313380	0.008906	2.701409	-0.462444	0.168810
pop	8.357304	8.268135	0.098324	-2.640056	0.715304	8.268135	8.435707
NX	0.078396	0.078396	0.133779	-0.258538	-0.053816	-0.020708	0.173023
public_spending	0.056326	0.056326	0.050162	0.580686	6.365286	0.021526	0.092522

Table 1: Descriptive statistics for Ireland (1996-2018)

	mean	median	std	skewness	kurtosis	25%	75%
patents	7.371374	7.282727	0.123304	-2.631118	0.884010	7.282727	7.439841
gdp	-3.688897	-0.283015	17.630964	-2.395712	7.000827	-0.504375	0.325560
pop	8.947566	8.897612	0.057785	-2.644053	0.621256	8.897612	8.992554
NX	0.073560	0.350255	1.131190	0.376465	2.915655	-0.074714	0.350255
public_spending	0.042721	0.042721	0.025264	0.746036	7.703924	0.026043	0.058422

Table 2: Descriptive statistics for Switzerland (1996-2018)

Patents (patents) : The average log of patents is significantly higher in Switzerland (7.37) than in Ireland (5.90), suggesting a more sustained innovation dynamic in Switzerland during the study period. The distribution of patent growth rates exhibits negative skewness in both countries, indicating a concentration of higher values, although the distributions remain relatively close to normality with moderate kurtosis.

GDP (gdp) : The average acceleration of GDP is negative in Switzerland (-3.69%) compared to a positive acceleration in Ireland (2.12%), highlighting significant differences in the economic growth dynamics of the two countries over the analyzed period. The distribution of GDP acceleration in Switzerland exhibits strong negative skewness and high kurtosis, suggesting a leptokurtic distribution with extreme variations, indicating significant instability in economic growth changes.

Population (pop) : The average population is higher in Switzerland (8.95) than in Ireland (8.36), indicating slightly higher population in Switzerland. Both distributions exhibit similar negative skewness, suggesting a homogeneous distribution around the median with a slight concentration of higher values.

(Net Trade Balance) (NX) : The average net trade balance growth rates are slightly higher in Ireland (7.84%) than in Switzerland (7.36%), although the differences are minimal. The distribution of NX growth rates in Ireland shows negative skewness, while in Switzerland it is slightly positive, indicating divergent trends in the trade performances of the two countries. The variability in kurtosis suggests differences in the frequency of extreme values between the two distributions.

Public Spending (public_spending) The average growth rates of public spending are higher in Ireland (5.63%) than in Switzerland (4.27%), which could reflect different budgetary priorities between the two nations. The distributions

of public spending growth rates show positive skewness in both countries, accompanied by high kurtosis, indicating a concentration of data around the median with heavy tails.

Data Limitations: While the descriptive statistics offer valuable insights into the growth rates of key economic variables for Ireland and Switzerland between 1996 and 2018, certain variables, such as GDP in Switzerland, exhibit highly skewed distributions and high kurtosis. These characteristics may indicate the presence of outliers or nonlinear behaviors, necessitating increased caution when interpreting the results and potentially justifying additional analyses to ensure the robustness of the conclusions.

The descriptive statistics in Table ^[1] and Table^[2] highlight both notable differences and significant similarities between Ireland and Switzerland. These commonalities reinforce the relevance of Switzerland as a control country for our study on the impact of the KDB reform in Ireland. These observations provide a solid foundation for the forthcoming econometric analysis, allowing for a rigorous evaluation of the reform's effects.

3.1.2 Synthetic control group

In addition to the DiD analysis, we applied the SCM to construct a more robust synthetic control group to evaluate the impact of the KDB reform in Ireland. This group primarily includes Austria, Finland, and Sweden, selected for their similarities with Ireland on key variables.

Selection and weighting of countries in the synthetic group: Austria, Finland, Sweden, and Switzerland were chosen due to their alignment with Ireland on significant economic and structural dimensions. These countries did not introduce reforms similar to Ireland's in 2015, making them appropriate for constituting a comparative synthetic group. Specifically, Switzerland introduced a patent box regime only in 2020 at the cantonal and communal levels²³, while the other countries do not have such incentives. The weights assigned to each country were optimized to maximize the resemblance between the synthetic group and Ireland before the introduction of the KDB reform. Other candidate countries, such as Portugal, were excluded due to missing data or divergent characteristics that could have compromised the robustness of the analysis.

	mean	median	std	skewness	kurtosis	25%	75%
patents	7.865349	7.490519	0.439371	-2.523629	0.898330	7.490519	8.340613
gdp	-0.101025	-0.101025	1.380291	-0.360537	2.162719	-0.525401	0.673947
pop	9.134493	9.095124	0.045016	-2.644691	0.594249	9.095124	9.165135
NX	-0.059303	0.046569	0.703159	0.288856	2.667265	-0.225358	0.046569
public_spending	0.030088	0.032175	0.021707	0.447375	7.438318	0.017464	0.044460

Table 3: Descriptive statistics for Sweden (1996-2018)

	mean	median	std	skewness	kurtosis	25%	75%
patents	7.594729	7.214871	0.450655	-2.406257	0.842438	7.214871	7.803788
gdp	-0.555866	-0.245525	2.642381	-1.810370	4.285757	-0.589991	0.297626
pop	8.576891	8.555316	0.024754	-2.645425	0.650580	8.555316	8.599047
NX	-0.381143	-0.084453	2.068574	-1.838769	4.094241	-0.323133	0.043308
public_spending	0.042183	0.042183	0.023885	0.614242	7.568300	0.028827	0.055595

Table 4: Descriptive statistics for Finland (1996-2018)

	mean	median	std	skewness	kurtosis	25%	75%
patents	8.960457	7.487912	1.220555	-1.945070	0.557468	7.487912	10.010883
gdp	-0.359839	-0.031561	1.248854	-1.750552	1.817735	-0.601065	0.281856
pop	9.023869	8.994940	0.032622	-2.645201	0.601198	8.994940	9.042127
NX	-0.743270	-0.124520	2.165271	-2.053429	4.170755	-0.493390	0.207698
public_spending	0.040725	0.041487	0.017502	0.321367	7.258828	0.024916	0.052467

Table 5: Descriptive statistics for Austria (1996-2018)

The descriptive statistics between Tables 3, 4, and 5 reveal notable differences and similarities among Finland, Sweden, and Austria. Sweden stands out with intermediate levels of patents and NX compared to Finland and Austria, while Austria shows significantly higher averages in terms of patents, reflecting a strong innovation dynamic. Differences in populations and public expenditures highlight varied socio-economic contexts, offering a rich perspective for analysis.

Despite these disparities, the similarities in key variables, such as the stability of macroeconomic indicators, reinforce the relevance of these countries in constructing a robust synthetic control group. These countries were selected because they did not introduce KDB-type reforms during the study period, thus justifying the use of Sweden, Finland, and Austria as the best available candidates to represent a balanced and pertinent control group.

3.2 Key variables for IP asset depreciation

The variables selected for the analysis aim to evaluate the impact of the KDB tax reform on the depreciation of IP assets. These variables, chosen for their economic and technological relevance, allow for a scientific exploration of the links between fiscal depreciation, innovation, and the management of intangible assets. They include:

Depreciation of IP Assets²⁴ (depreciation), (OECD): Calculated as the difference between gross assets and net assets, this indicator reflects the wear or amortization of intangible assets, a key element in the analysis of innovation and fiscal policies. *Data transformation: in growth rate (%)*

Average Wages²⁵ (salary) in US dollars, PPP (Purchasing power parity) converted base 2022, (OECD): An essential economic and social indicator for assessing purchasing power and living conditions. *Data transformation: in growth rate (%)*

Unemployment²⁶ (cho) as % of the population, (World Bank): A key labor market indicator reflecting the proportion of unemployed individuals within the population. *Data transformation: in growth rate (%)*

Purchasing Power Parity²² (ppa) in international dollars, (OECD): A comparative measure for evaluating the relative cost of living and purchasing power between countries. *Data transformation: in growth rate (%)*

Foreign Direct Investment²⁷ (IDE) in billions of dollars, (World Bank): An indicator of foreign capital flows, crucial for understanding a country's attractiveness and international integration. *Data transformation: in growth rate (%)*

3.2.1 Descriptive statistics

	mean	median	std	skewness	kurtosis	25%	75%
depreciation	0.214165	0.214165	0.140926	2.029147	6.139046	0.149481	0.222253
salary	0.023136	0.027651	0.027651	0.970080	7.755251	0.005670	0.038785
cho	-0.008573	0.059282	0.243638	1.724398	2.544239	-0.150826	0.059282
ppa	-0.000130	-0.000130	0.028048	-0.326634	6.215840	-0.016704	0.021526
IDE	0.087292	0.087292	1.526163	-1.427166	2.265027	-0.534993	1.018203

Table 6: Descriptive statistics for Ireland (1996-2018)

	mean	median	std	skewness	kurtosis	25%	75%
depreciation	0.061280	0.061280	0.049087	-0.168794	2.852770	0.027157	0.103613
salary	0.010804	0.010860	0.010860	0.160362	6.810873	0.003050	0.019853
cho	-0.031283	0.008620	0.098811	1.013048	4.885598	-0.081794	0.008620
ppa	-0.006140	-0.002847	0.013575	-0.322323	7.544923	-0.018134	0.003011
IDE	-0.198059	-0.198059	1.887312	-0.374108	1.323445	-1.478749	0.702093

Table 7: Descriptive statistics for Finland (1996-2018)

Depreciation of IP Assets²⁴ (depreciation) : The average growth rate of asset depreciation is significantly higher in Ireland (0.214) than in Finland (0.061). This difference could reflect distinct accounting or fiscal practices in managing

intangible assets between the two countries. The distribution of depreciation rates in Ireland shows marked positive skewness (skewness = 2.029) and high kurtosis (6.139), indicating a concentration of high values and the presence of extreme behaviors. In Finland, the distribution is more symmetrical (skewness = -0.169) with moderate kurtosis (2.853), suggesting a more homogeneous distribution of depreciation rates.

Average Wages²⁵ (salary) : The average growth rate of wages is higher in Ireland (0.023) than in Finland (0.011), suggesting a more robust wage dynamic in Ireland. The wage distribution in Ireland shows positive skewness (skewness = 0.970) and very high kurtosis (7.755), which may indicate increased variability and the presence of extreme values. In Finland, the wage distribution is relatively symmetrical (skewness = 0.160) with high kurtosis (6.811), reflecting relative stability in wage growth.

Unemployment²⁶ (cho) : The average unemployment growth rate is slightly negative in Ireland (-0.009) and more pronounced in Finland (-0.031), indicating a more marked improvement in the labor market in Finland. The distributions of unemployment growth rates show positive skewness in Ireland (skewness = 1.724) and Finland (skewness = 1.013), accompanied by high kurtosis (2.544 in Ireland and 4.886 in Finland). This suggests a concentration of values around the median with the presence of extreme values influencing the distribution.

Purchasing Power Parity²² (ppa) : The average PPP growth rate is almost zero in Ireland (-0.00013) and slightly negative in Finland (-0.00614), indicating relative stability in international comparisons based on this indicator in both countries. The distributions show slight negative skewness (skewness = -0.327 for Ireland and -0.322 for Finland) and high kurtosis (6.216 in Ireland and 7.545 in Finland), suggesting a concentrated distribution of values with some extremes.

Foreign Direct Investment²⁷ (IDE) : FDI shows positive average growth in Ireland (0.087) versus negative growth in Finland (-0.198), illustrating a much higher international attractiveness for Ireland. The FDI distribution in Ireland exhibits marked negative skewness (skewness = -1.427) and moderate kurtosis (2.265), indicating a concentration of low values with some high extreme values. In Finland, the FDI distribution is also negatively skewed (skewness = -0.374) but with very low kurtosis (1.323), suggesting a flatter distribution of values without marked concentration of extremes.

Data limitations: While the descriptive statistics provide a relevant overview of the growth rates of key economic variables for Ireland and Finland between 1996 and 2018, the sample size and the presence of highly skewed and leptokurtic distributions may limit the generalization of the results.

3.3 SCM Descriptive statistics

	mean	median	std	skewness	kurtosis	25%	75%
depreciation	0.067236	0.066536	0.012864	-0.821822	7.856852	0.059778	0.074699
salary	0.006556	0.010558	0.010558	0.568990	7.908582	0.000287	0.015065
cho	0.012732	0.061906	0.127048	0.688902	3.054022	-0.080102	0.061906
ppa	-0.008751	-0.000832	0.011604	0.183502	6.947915	-0.016686	-0.000832
IDE	2.309527	2.070032	9.042014	2.070032	5.573037	-0.947667	1.793902

Table 8: Descriptive statistics for Austria (1996-2018)

The descriptive statistics in Table^[8] highlight the notable stability of asset depreciation in Austria, with low variability and a negatively skewed distribution. Wages and FDI exhibit specific dynamics, reflecting distinct economic behaviors. PPP remains stable, reinforcing Austria's relevance as a candidate for the synthetic control group.

The absence of complete data for certain variables in other countries like Sweden and Switzerland justifies their exclusion, thus reinforcing the methodological approach adopted for constructing the synthetic control group.

Building on this presentation of the data, we now move on to the description of the empirical strategy implemented to assess the impact of the reform.

4 Empirical strategy

This section describes the methodology employed to evaluate the impact of the introduction of the KDB in Ireland on the number of patents filed and the depreciation rate of IP assets. We used a combined approach to ensure the robustness of our results, including a DiD analysis and a dynamic DiD analysis (event study) with Ireland as the treated country. For the patent analysis, Switzerland is used as the control country, while for the depreciation of IP assets, Finland serves as the control country. Additionally, we applied a SCM including Finland, Switzerland, Austria, and Sweden for the patent analysis, and an SCM with Austria and Finland for the depreciation of IP assets due to limitations in available data. We also used a linear trend model to test the parallel trends assumption, essential to the validity of the DiD and SCM methods. The selection of these countries allows for the creation of reliable comparison groups despite observed differences. In this section, we will detail the econometric models used, the underlying assumptions, as well as the techniques for validating the results.

4.1 Linear trend model

To test the parallel trends, we use the model available in the `statsmodel`⁶ package in *Python*, which adds interaction terms:

$$y_{ist} = DID_{ist} + w_i d_{t,0} \zeta_1 + w_i d_{t,1} \zeta_2 + \epsilon_{ist}$$

where $d_{t,0}$ and $d_{t,1}$ indicate the pre- and post-treatment periods, and w_i indicates the treated group. The Wald test evaluates whether $\zeta_1 = 0$, verifying the parallel trends in pre-treatment.

The null hypothesis (H_0) is that the trends are parallel. Rejection of H_0 indicates a violation of parallel trends, which calls into question the validity of the DiD results. Our results show a significant divergence, indicating a violation of parallel trends.

4.2 Difference-in-Differences

The DiD method compares the observed changes before and after the reform, on one hand for patents in Ireland compared to Switzerland, and on the other hand for the depreciation of IP assets in Ireland compared to Finland. To perform the analysis, we use the `feols`²⁸ package available in *R*. The DiD model is specified as follows:

4.2.1 Model 1: Patents

$$\log(\text{Patents}_{it}) = \alpha + \tau(\text{Treatment}_i \times \text{Post}_t) + \sum_{k=1}^4 \beta_k \mathbf{X}_{kit} + \mu_i + \delta_t + \epsilon_{it}$$

Symbol	Description	Indexing
$\log(\text{Patents}_{it})$	log of patents for country i in year t .	-
α	Intercept.	-
τ	Coefficient associated with the interaction $\text{Treatment}_i \times \text{Post}_t$.	-
β_k	Estimated coefficient for variable $\mathbf{X}_k$ for country i in year t	$k = 1, \dots, 4$
Treatment_i	Indicator variable: 1 for the treated group, 0 otherwise.	-
$\mathbf{X}_{kit}$	Control variables:	$k = 1, \dots, 4$
<i>Details of the control variables $\mathbf{X}_{kit}$:</i>		
GDP_{it}	Δ^2 Growth rate of the grow rate of the real GDP (acceleration).	$k = 1$
Population_{it}	log of population.	$k = 2$
NX_{it}	Growth rate (%) of net exports.	$k = 3$
$\text{PublicSpending}_{it}$	Growth rate (%) of public spending.	$k = 4$
μ_i	Country-specific fixed effects for country i .	-
δ_t	Time fixed effects for year t .	-
ϵ_{it}	Random error term.	-

Table 9: Description of the variables used in the model for patents

4.2.2 Model 2: Depreciation of IP Assets

$$\Delta\text{Depreciation}_{it} = \alpha + \tau(\text{Treatment}_i \times \text{Post}_t) + \sum_{k=1}^4 \beta_k \mathbf{Z}_{kit} + \mu_i + \delta_t + \epsilon_{it}$$

Symbol	Description	Indexing
$\Delta\text{Depreciation}_{it}$	Growth rate (%) Depreciation rate of IP assets for country i in year t .	-
α	Intercept.	-
τ	Coefficient associated with the interaction $\text{Treatment}_i \times \text{Post}_t$.	-
β_k	Estimated coefficient for variable $\mathbf{Z}_k$ for country i in year t	$k = 1, \dots, 4$
Treatment_i	Indicator variable: 1 for the treated group, 0 otherwise.	-
Post_t	Indicator variable: 1 for post-reform periods, 0 otherwise.	-
$\mathbf{Z}_{kit}$	Control variables:	$k = 1, \dots, 4$
<i>Details of the control variables $\mathbf{Z}_{kit}$:</i>		
salary_{it}	Growth rate (%) of salaries.	$k = 1$
cho_{it}	Growth rate (%) of unemployment.	$k = 2$
ppa_{it}	Growth rate (%) of Purchasing Power Parity.	$k = 3$
IDE_{it}	Growth rate (%) of Foreign Direct Investments.	$k = 4$
μ_i	Country-specific fixed effects for country i .	-
δ_t	Time fixed effects for year t .	-
ϵ_{it}	Random error term.	-

Table 10: Description of the variables used in the model of IP assets depreciation

4.2.3 DiD model assumptions

In this study covering the period from 1996 to 2018, we applied the DiD model to evaluate the impact of the reform on two main measures: the number of patents and the growth rate of depreciation of IP assets. Ireland is the treated country, while Switzerland serves as the control group for the patent analysis and Finland for the depreciation of IP assets analysis. The reform was voted in 2015 and took effect in 2015. The following assumptions underlie the validity of our DiD approach:

Parallel trends assumption: Before the introduction of the reform in 2015, the trends in the number of patents in Ireland (treated) and Switzerland (control group), as well as the growth rates of depreciation of IP assets in Ireland and Finland (control group), evolved in a parallel manner. This assumption ensures that, in the absence of intervention, both measures would have followed similar trajectories in the respective treated and control countries.

No interference between units: The reform in Ireland does not directly affect the number of patents or the depreciation of IP assets in the control countries (Switzerland for patents and Finland for depreciation), thereby avoiding any contamination or interference between the treated and control groups.

Stability of control variable effects: The coefficients of the control variables ($\mathbf{X}_{kit}$ for the patent model and $\mathbf{Z}_{kit}$ for the depreciation of IP assets model) are constant over the study period. This implies that the impact of these variables on the dependent measures remains stable over time.

Exogeneity of explanatory variables: The explanatory variables used in both models ($\mathbf{X}_{kit}$ and $\mathbf{Z}_{kit}$) are not correlated with the error term ϵ_{it} . This condition ensures that the control variables correctly capture the factors influencing the dependent measures without introducing bias in the coefficient estimates.

No anticipation effects: The units (Ireland, Switzerland, and Finland) do not alter their behaviors in anticipation of the reform before its introduction in 2015. This means that the trends observed before the reform are not influenced by anticipations of change.

4.3 Dynamic Difference-in-Differences (Event study)

This method involves estimating the effects of the reform at different time horizons, thereby allowing the visualization of the evolution of the impact over time. To do so, we include indicator variables for each period relative to the event (reform), typically centered around the treatment period. This allows for the detection of anticipated effects before the reform and the measurement of the magnitude and persistence of the effects after the reform. To perform the analysis, we use the `feols`²⁸ package available in *R*

4.3.1 Dynamic model specification

The dynamic DiD model²⁹ can be specified as follows for the two variables we measure:

$$\log(\text{Patent}_{it}) = \alpha + \sum_{m=-M}^N \gamma_m \text{Post}_{mt} + \sum_{k=1}^4 \beta_k \mathbf{X}_{kit} + \mu_i + \delta_t + \epsilon_{it}$$

$$\Delta \text{Depreciation}_{it} = \alpha + \sum_{m=-M}^N \gamma_m \text{Post}_{mt} + \sum_{k=1}^4 \beta_k \mathbf{Z}_{kit} + \mu_i + \delta_t + \epsilon_{it}$$

Model 2: Depreciation of IP assets

- Post_{mt} represents the indicator variables for each period m relative to the reform (with $m = 0$ corresponding to the reform period).
- M is the number of periods before the reform and N is the number of periods after the reform.
- The other terms are defined as in the main DiD model^(4.2).
- For $m < 0$, the coefficients indicate whether there were pre-existing trends before the reform.
- For $m \geq 0$, they measure the impact of the reform on the number of patents in each post-reform period.

Robustness tests

To ensure the robustness of the results, we perform the following tests:

- **Parallel Trends**^[4.1]: Verify that the trends in patents were similar between the treated and control groups before the reform.
- **Dynamic Fixed Effects**: Include fixed effects to capture unobserved variations specific to countries and periods.
- **Control for Time and Country Effects**: Ensure that global time effects and country-specific characteristics are properly controlled.

Visualization of dynamic effects

We present the coefficients γ_m in the form of event study graphs, illustrating the effects of the reform on the number of patents before and after the intervention. These visualizations allow for the visual detection of temporal variations and the assessment of the persistence of effects.

4.4 Synthetic control method (SCM)³⁰

The SCM is used to create synthetic control groups that match the characteristics of Ireland before the introduction of the reform. For the patent model, SCM combines data from Austria, Switzerland, Finland, and Sweden to minimize differences

with Ireland on the control variables. In contrast, for the depreciation of IP assets model, only Austria and Finland are used as synthetic control countries due to data limitations. By assigning optimal weights to the countries in each synthetic control group, SCM seeks to create linear combinations of the control countries that best mimic the characteristics of Ireland before the implementation of the reform. The weights are determined such that the sum of the weights equals one and no weight is negative, thereby ensuring a faithful and balanced representation of Ireland. To perform the analysis, we use the `synthdid`³¹ package available on *github.com* in *Python*.

4.4.1 Patent model

$$\min_{\omega} \left(\mathbf{X}_{\text{IRL}} - [\mathbf{X}_{\text{AUT}}, \mathbf{X}_{\text{CHE}}, \mathbf{X}_{\text{FIN}}, \mathbf{X}_{\text{SWE}}] \begin{bmatrix} \omega_{\text{AUT}} \\ \omega_{\text{CHE}} \\ \omega_{\text{FIN}} \\ \omega_{\text{SWE}} \end{bmatrix} \right)^T \mathbf{V} \left(\mathbf{X}_{\text{IRL}} - [\mathbf{X}_{\text{AUT}}, \mathbf{X}_{\text{CHE}}, \mathbf{X}_{\text{FIN}}, \mathbf{X}_{\text{SWE}}] \begin{bmatrix} \omega_{\text{AUT}} \\ \omega_{\text{CHE}} \\ \omega_{\text{FIN}} \\ \omega_{\text{SWE}} \end{bmatrix} \right) \quad \text{s.t.} \begin{cases} \sum \omega_k = 1 \\ \omega_k \geq 0 \end{cases}$$

Symbol	Description	Indexing
$\mathbf{X}_{\text{IRL}}$	Vector of control variables for Ireland.	-
$\mathbf{X}_{\text{AUT}}$	Vector of control variables for Austria.	$k = 1$
$\mathbf{X}_{\text{CHE}}$	Vector of control variables for Switzerland.	$k = 2$
$\mathbf{X}_{\text{FIN}}$	Vector of control variables for Finland.	$k = 3$
$\mathbf{X}_{\text{SWE}}$	Vector of control variables for Sweden.	$k = 4$
ω_{AUT}	Weight for Austria in the synthetic control group.	-
ω_{CHE}	Weight for Switzerland in the synthetic control group.	-
ω_{FIN}	Weight for Finland in the synthetic control group.	-
ω_{SWE}	Weight for Sweden in the synthetic control group.	-
$\mathbf{V}$	Positive semi-definite weighting matrix.	-

Table 11: Description of the variables used in the Patent Model

The estimation of synthetic patents for Ireland is given by:

$$\hat{\mathbf{y}}_{\text{IRL},t}^{\text{Synth}} = \sum_{k \in \{\text{AUT}, \text{CHE}, \text{FIN}, \text{SWE}\}} \omega_k \cdot \mathbf{y}_{k,t}$$

The measure of the reform's effect on patents is the difference between the observed number of patents in Ireland and those estimated for the synthetic control group:

$$\Delta \mathbf{y}_{\text{IRL},t} = \mathbf{y}_{\text{IRL},t} - \hat{\mathbf{y}}_{\text{IRL},t}^{\text{Synth}}$$

The root mean squared prediction error (RMSPE) is used to evaluate the quality of the model fit:

$$\text{RMSPE} = \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{\mathbf{y}}_{\text{synthetic},t} - \mathbf{y}_{\text{treated},t})^2}$$

4.4.2 Depreciation of IP Assets Model

$$\min_{\omega} \left(\mathbf{X}_{\text{IRL}} - [\mathbf{X}_{\text{AUT}}, \mathbf{X}_{\text{FIN}}] \begin{bmatrix} \omega_{\text{AUT}} \\ \omega_{\text{FIN}} \end{bmatrix} \right)^T \mathbf{V} \left(\mathbf{X}_{\text{IRL}} - [\mathbf{X}_{\text{AUT}}, \mathbf{X}_{\text{FIN}}] \begin{bmatrix} \omega_{\text{AUT}} \\ \omega_{\text{FIN}} \end{bmatrix} \right) \quad \text{s.t.} \begin{cases} \sum \omega_k = 1 \\ \omega_k \geq 0 \end{cases}$$

The estimation of synthetic depreciation of IP assets for Ireland is given by:

$$\hat{\mathbf{y}}_{\text{IRL},t}^{\text{Synth}} = \sum_{k \in \{\text{AUT}, \text{FIN}\}} \omega_k \cdot \mathbf{y}_{k,t}$$

Symbol	Description	Indexing
$\mathbf{X}_{\text{IRL}}$	Vector of control variables for Ireland.	-
$\mathbf{X}_{\text{AUT}}$	Vector of control variables for Austria.	$k = 1$
$\mathbf{X}_{\text{FIN}}$	Vector of control variables for Finland.	$k = 2$
ω_{AUT}	Weight for Austria in the synthetic control group.	-
ω_{FIN}	Weight for Finland in the synthetic control group.	-
$\mathbf{V}$	Positive semi-definite weighting matrix.	-

Table 12: Description of the variables used in the Depreciation of IP Assets Model

The measure of the reform’s effect on the depreciation of IP assets is the difference between the observed depreciation in Ireland and that estimated for the synthetic control group:

$$\Delta \mathbf{y}_{\text{IRL},t} = \mathbf{y}_{\text{IRL},t} - \hat{\mathbf{y}}_{\text{IRL},t}^{\text{Synth}}$$

The root mean squared prediction error (RMSPE) is used to evaluate the quality of the model fit:

$$\text{RMSPE} = \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{y}_{\text{synthetic},t} - y_{\text{treated},t})^2}$$

4.4.3 SCM Model assumptions

Convex weights: The weights must be non-negative and their sum must equal one.

Pre-treatment similarity: Control units must have characteristics and trends similar to those of the treated unit before the intervention.

Stability of covariate effects: The relationships between the covariates and the outcome of interest must remain stable over time.

Exclusion of Significant Disturbances: No other major events should influence the variables of interest during the study period.

Conditional Independence: Conditional on the covariates, the outcomes of the treated unit and the control units are independent.

Having presented the data utilized and laid out the methodological framework, we now turn to the analysis of results to assess the impact of the reform.

5 Results

This section presents the results of our empirical analysis aimed at evaluating the impact of the reform in Ireland on two key indicators: the number of patents and the growth rate of the depreciation of IP assets. To ensure the robustness of our conclusions, we first used the linear trends model ^{[4.1]6}, we then employed three complementary approaches: the DiD method, the dynamic DiD method (event study), and the SCM. The results obtained from these methods are discussed in detail in the following subsections, highlighting the effects of the reform on each indicator.

5.1 Linear trends

5.1.1 Patents

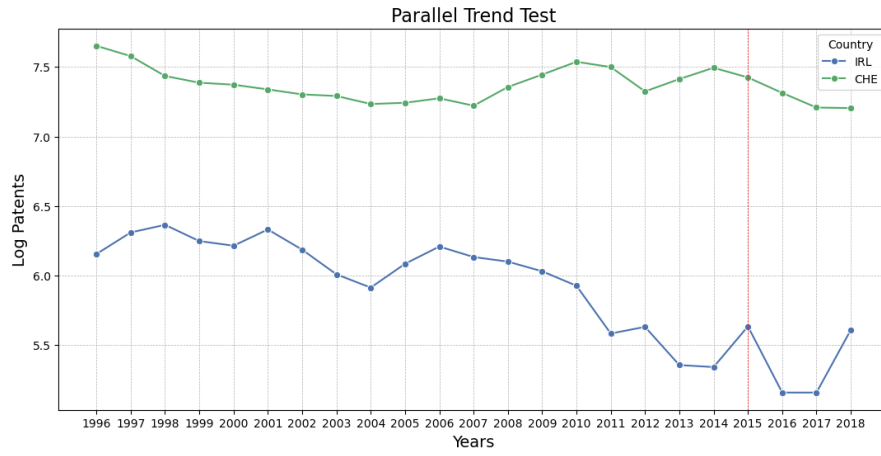


Figure 1

In addition to the graph, we present Table ^[13] below, containing the results of the statistical test for parallel trends. The results indicate that the trends in the number of patents in Ireland (treated) and Switzerland (control) are not perfectly parallel during the pre-treatment period. This lack of parallelism is evidenced by an extremely low p-value (7.23×10^{-11}), associated with an F-statistic of 31.93, which leads to a significant rejection of the null hypothesis (H_0 : the trends are parallel). These results suggest a significant violation of the parallel trends assumption^[4.2.3], necessitating increased caution in interpreting the results of the DiD and dynamic DiD approaches.

N	F-statistic	Prob (F-statistic)
46	31.93	7.23e-11

Table 13: Results of the parallel trends test^[Appendix 23]

5.1.2 Growth Rate of Depreciation of Current Intellectual Property Assets

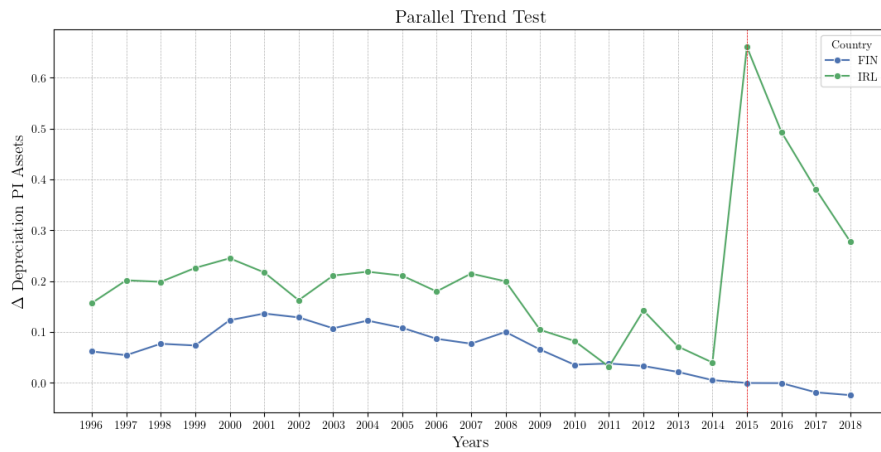


Figure 2

Similarly, we present Table^[24] below, containing the results of the statistical test for parallel trends. The results show that the trends in the growth rate of the depreciation of IP assets between Ireland (treated) and Finland (control) exhibit considerable divergence during the pre-treatment period. This is evidenced by an F-statistic of 47.25 and a p-value of 2.02e-12, indicating a rejection of the null hypothesis of parallel trends. This suggests caution in interpreting the results of the DiD approach^[5,2] and the dynamic DiD, particularly for these measures.

<i>N</i>	F-statistic	Prob (F-statistic)
46	47.25	2.02e-12

Table 14: Results of the parallel trends test^[Appendix 24]

5.2 Difference-In-Differences and Dynamic DiD (Event study)

5.2.1 Static DiD for Patents

The static DiD analysis shows a significant decrease in the number of patents post-reform. However, this result should be interpreted with caution due to the violation of the parallel trends.

Dependent Variable:	patents
Model:	(1)
<i>Variables</i>	
did	-0.2766*** (9.69×10^{-16})
gdp	-0.0025*** (1.24×10^{-17})
pop	-3.236*** (3.84×10^{-14})
NX	-0.0188*** (3.96×10^{-16})
public_spending	5.242*** (1.61×10^{-14})
<i>Fixed-effects</i>	
country	Yes
year	Yes
<i>Fit statistics</i>	
Observations	46
R ²	0.98654
Within R ²	0.76493
<i>Clustered (country) standard-errors in parentheses</i>	
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>	

Table 15: DiD Results on patents

- The analysis of the results in Table^[15] highlights a significant average decrease in the number of patents attributable to the effect estimated by the DiD method. The main variable of interest, did, shows a coefficient of -0.2766, highly significant ($p \approx 0$). This result indicates a notable reduction in patents in the treated countries after the implementation of the reform.
- The other variables in the model also provide relevant information. For example, GDP (gdp) has a significant negative effect, suggesting that higher levels of wealth are associated with a marginal decrease in the number of patents. Similarly, population (pop) and net export balance (NX) also have significant negative effects on the number of patents, while public spending (public_spending) has a marked positive impact.
- With an $R^2 = 0.98654$ and a within R^2 of 0.76493, the model captures a substantial proportion of the variance in patents, reflecting a very good fit. Moreover, the inclusion of fixed effects by country and year reinforces the relevance of the approach used to account for temporal and structural specificities.

The decrease in patents may indicate that the tax reform encouraged companies to locate their IP income in Ireland without necessarily stimulating genuine innovation. This raises questions about the effectiveness of patent boxes in attracting R&D investments. For investors, this could signal limited growth opportunities in sectors dependent on local innovation.

Finally, although these results are robust in terms of statistical significance, they should be interpreted with caution due to the violation of the parallel trends assumption^[13] during the pre-treatment period; these results do not allow for a definitive conclusion on a causal effect. Despite this limitation, they provide interesting elements to better understand the potential impact of the reform on innovation activity, while opening perspectives for complementary research to confirm these observations.

5.2.2 Dynamic DiD for Patents

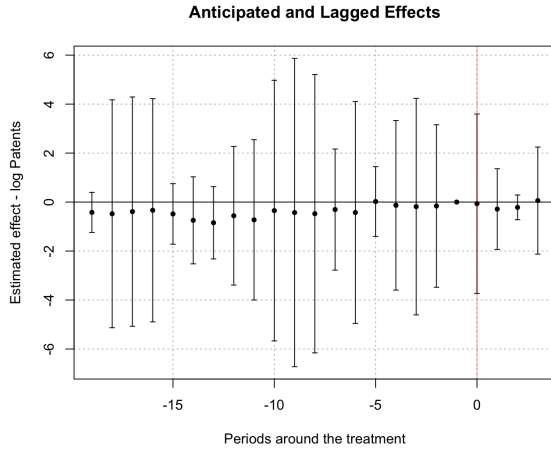


Figure 3: Graph from Table ^[25]

- Figure ^[3] is the result of the event study (Table ^[25]). Its analysis complements the results of the static DiD model (Table ^[15]) by providing information on the dynamics of effects before and after the reform. Before the treatment (negative periods), some coefficients are significant, notably at -19 (-0.4217^*) and -13 (-0.8450^*). These results highlight a significant decrease in patents in certain pre-treatment periods, which constitutes a violation of the parallel trends assumption. This limitation, already discussed in the static model, prevents fully validating the causal interpretation.

Around year 0, the estimated coefficients are close to zero and not significant (e.g., -0.0654 for $t = 0$), suggesting an absence of immediate effect of the reform on the number of patents. However, in the post-treatment periods, a significant decrease appears at $t = 2$ (-0.2163^{***}). This result is consistent with the negative coefficient obtained in the static model ($did = -0.2766^{***}$) and indicates a delayed effect of the reform on innovation. The Event Study model also highlights robust effects for the explanatory variables. For example, GDP (gdp) and population (pop) have significant negative effects (-0.0026^{***} and -3.912^{***} , respectively), indicating their restrictive role on the number of patents. Similarly, the net trade balance (NX) has a significant negative effect (-0.0303^{***}), while public spending (public_spending) has a significant positive impact (5.724^{***}), highlighting their contribution to supporting innovation. The dynamic analysis reveals a significant decrease in patents starting from $t = 2$ (-0.2163), indicating a delayed effect of the reform, consistent with the results of the static model (-0.2766). However, the violations of parallel trends in the pre-treatment period limit the causal validity of the conclusions and underline the need for complementary models to confirm these observations.

Nevertheless, these results support the work of Bloom et al. (2002)⁸, who show that tax incentives often require several years to produce a significant impact on innovation, reflecting a latency in companies' strategic responses to reforms.

5.2.3 Static DiD for the Depreciation Rate of Intellectual Property Assets

The static DiD analysis shows a significant increase in the depreciation rate of IP assets post-reform. However, this result should be interpreted with caution due to the violation of the parallel trends.

Dependent Variable:	patents
Model:	(1)
<i>Variables</i>	
did	-0.2766*** (9.69×10^{-16})
gdp	-0.0025*** (1.24×10^{-17})
pop	-3.236*** (3.84×10^{-14})
NX	-0.0188*** (3.96×10^{-16})
public_spending	5.242*** (1.61×10^{-14})
<i>Fixed-effects</i>	
country	Yes
year	Yes
<i>Fit statistics</i>	
Observations	46
R ²	0.98654
Within R ²	0.76493
<i>Clustered (country) standard-errors in parentheses</i>	
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>	

Table 16: DiD Results for depreciation of IP assets

- The analysis of Table ^[16] reveals a significant decrease in the number of patents attributable to the studied reform. The main variable did presents an estimated coefficient of -0.2766, highly significant ($p \approx 0$), indicating a notable reduction in innovation activity after the introduction of the reform in the treated countries.
- Among the other explanatory variables, GDP (gdp) and population (pop) display significant negative effects, indicating their role in reducing the number of patents. Net export balance (NX) follows the same negative trend, while public spending (public_spending) exerts a positive and significant effect, highlighting their importance in supporting innovation.
- The overall model shows excellent explanatory capacity, with an $R^2 = 0.98654$ and a within R^2 of 0.76493, indicating that nearly 98.6% of the variance in patents is explained by the included variables. Fixed effects by country and year ensure that structural and temporal specificities are well accounted for in the analysis.

The increase in the depreciation rate of IP assets may indicate that companies are valuing these assets more in their balance sheets due to the tax reform. This could signal an accounting strategy aimed at optimizing tax advantages, without necessarily reflecting a real increase in innovation. For investors, an increase in depreciation could reduce visibility on the true economic performance of companies, making investment decisions more uncertain. Companies, by adjusting their accounting management, could reduce their tax burden in the short term but risk a loss of confidence in the long term if these adjustments are not supported by real innovations. These results corroborate the observations of Alstadsaeter et al. (2018)¹⁴, who note that patent boxes may encourage tax strategies rather than a significant increase in R&D investments.

Finally, although these results are robust in terms of statistical significance, they should be interpreted with caution due to the violation of the parallel trends assumption ^[24] during the pre-treatment period; these results do not allow us to conclude a causal effect. Despite this limitation, they provide interesting elements to better understand the potential impact of the reform on innovation activity.

5.2.4 Dynamic DiD for the depreciation rate of IP assets

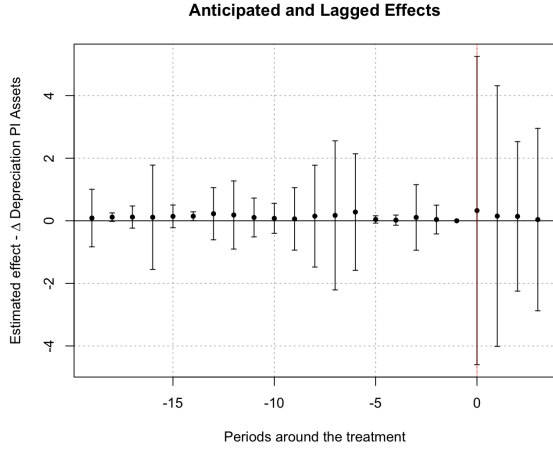


Figure 4: Graph from Table [26]

- Figure [4] is the result of the event study (Table [26]). Its analysis complements the results of the static DiD model (Table [16]) by providing information on the dynamics of effects before and after the reform. Before the treatment (negative periods), several coefficients are significant, notably at -18 (0.1165*) and -14 (0.1459**). These significant coefficients in the pre-treatment period question the validity of the parallel trends assumption, already identified as a limitation in the static analysis.
- Around year 0, the estimated coefficient (0.3267) is not significant, indicating an absence of immediate effect of the reform on the depreciation rate of IP assets. In the post-treatment periods (1, 2, and 3), the coefficients also remain non-significant, suggesting that the reform did not produce a notable delayed effect on the depreciation rate.
- The event study model also highlights the robustness of the effects of the main explanatory variables. For example, salary (salary) shows a positive and highly significant effect (0.9891***), reflecting its key role in the dynamics of IP assets. In contrast, the unemployment rate (cho) and asset prices (ppa) have significant negative effects (−0.4850*** and −1.722***, respectively), which is consistent with economic expectations. Finally, foreign direct investment (IDE) also shows a significant negative effect (−0.0168***), reinforcing the idea of a complex relationship between investment rates and the depreciation rate.

The dynamic analysis does not indicate any significant delayed effect of the reform on the depreciation rate of IP assets. Although significant coefficients are observed before treatment ($t = -18, t = -14$), questioning the validity of parallel trends, the post-treatment periods (1, 2, and 3) show coefficients close to zero. These results suggest that the reform did not significantly alter the valuation or management of IP assets. The absence of a delayed effect for the depreciation of IP assets could reflect inertia of companies in the face of tax incentives or strategic choices limited by other constraints. These results align with the work of Alstadsaeter et al. (2018)¹⁴, who show that patent boxes often encourage fiscal adjustments without necessarily stimulating genuine innovation. A comparison with other similar studies could, however, strengthen the robustness of these conclusions.

In conclusion, these results suggest that the reform did not significantly alter the valuation or management of IP assets.

5.3 Results of the synthetic control method (SCM)

In this section, we present the results of our analysis based on the SCM^[4,4]. This approach allows us to construct a synthetic control group that reproduces the characteristics of Ireland before the introduction of the 2015 tax reform. For the analysis related to patents, we selected Switzerland, Finland, Austria, and Sweden as control countries due to their similarities with Ireland on key variables. Regarding the depreciation of IP assets, only Finland and Austria were retained to constitute the synthetic group due to data availability.

5.3.1 Patents

The analysis of Table^[17] shows that the synthetic control relies exclusively on Switzerland (weight ≈ 1.00) to represent the counterfactual group of Ireland, due to the close correspondence between their dynamics of the logarithm of patents in the pre-treatment period. This selection is consistent with the parallel trends Figure^[1] and supports the relevance of Switzerland as the best control for the static DiD. The RMSPE score (1.8937) is relatively high and reflects significant variability, indicating that the synthetic control does not perfectly adjust the pre-treatment levels. Despite this, the estimated effect ($\hat{\tau} = -1.93254$) indicates a notable decrease in the logarithm of patents after the reform, consistent with the DiD results ($\text{did} = -0.2766^{***}$).

However, the violation of parallel trends in the static analysis limits the validity of causal conclusions. The total dependence on Switzerland in the synthetic control must also be considered an important methodological constraint.

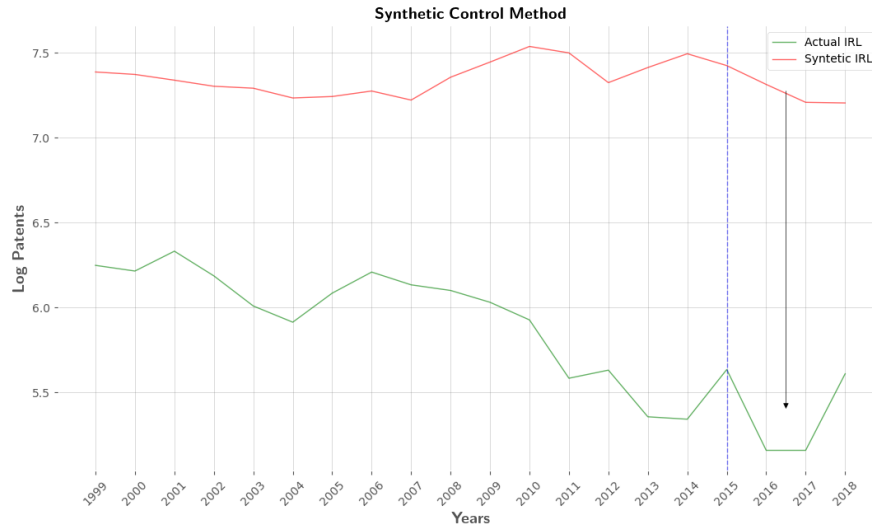


Figure 5

	Country Weights for the Synthetic Control				Performance Indicators	
	Austria	Finland	Switzerland	Sweden	RMSPE	$\hat{\tau}$
weights	3.385307×10^{-15}	0.00	1.00	2.621472×10^{-15}	1.8937	-1.93254

Table 17: Summary Table of results: Country weights for the synthetic control and results

5.3.2 Growth rate of depreciation of current IP assets

The analysis of Table^[18] shows that the synthetic control relies exclusively on Finland (weight = 1.00) to represent the counterfactual group of Ireland. This weighting indicates that, among the available countries, Finland best matches Ireland on the control variables related to the growth rate of the depreciation of IP assets. This result is consistent with the parallel trends Figure^[2], where the trajectories of Finland and Ireland appear closely aligned before the reform, as also reflected in Figure^[6]. The RMSPE (0.227972) reflects a variability of 23% between the observed and synthetic values before the reform. Although relatively high for a growth rate, it indicates a reasonable, but not perfect, correspondence between Ireland and its synthetic control. The effect estimated by the SCM ($\hat{\tau} = 0.39848$) suggests a moderate increase in the growth rate of IP asset depreciation post-reform, in line with the static DiD analysis.

However, the violation of parallel trends identified in the preliminary analyses limits the robustness of causal conclusions.

Furthermore, the exclusive dependence on Finland in the synthetic control reduces the diversity of the counterfactual group, which may influence the precision of the estimates. These observations, while interesting, must be interpreted with caution in evaluating the impact of the reform on IP assets.

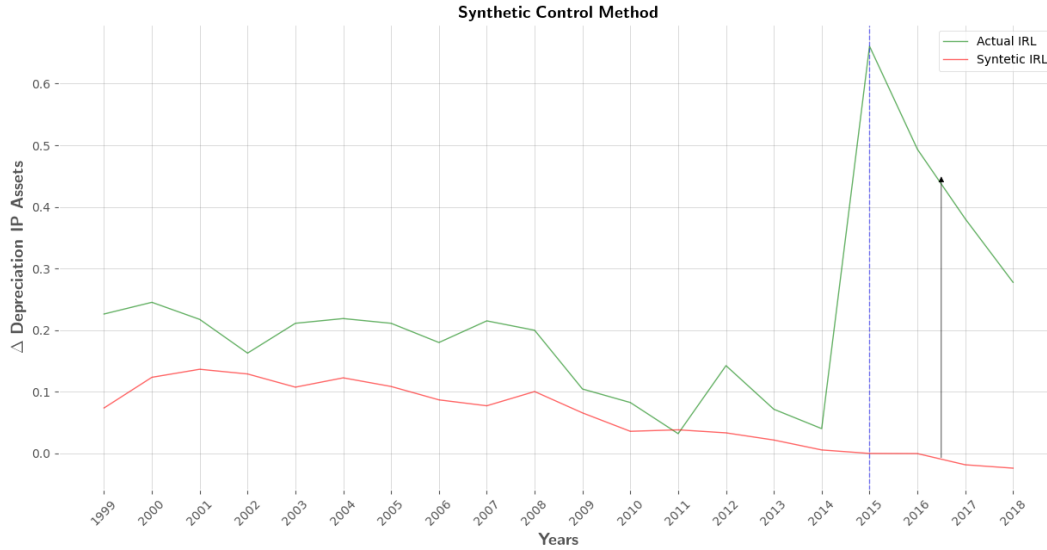


Figure 6

Country Weights for the Synthetic Control			Performance Indicators	
	Austria	Finland	RMSPE	$\hat{\tau}$
weights	0.00	1.00	0.227972	0.39848

Table 18: Country weights for the synthetic control and performance indicators for the growth rate of IP asset depreciation

5.4 Interpretation of the results

The results of our empirical analysis provide significant indications on the impact of the tax reform in Ireland on innovation activity, measured by the number of patents, and on the management of IP assets, evaluated by the growth rate of the depreciation of these assets. Despite the methodological limitations related to the violation of the parallel trends assumption, our different analytical approaches converge towards consistent conclusions, suggesting notable effects of the reform. In the case of the number of patents, the results of the static DiD model indicate a significant average decrease post-reform. This decline is confirmed by the event study analysis (Dynamic DiD), which reveals a significant decrease in patents two years after the implementation of the reform. Moreover, the SCM reinforces this observation by estimating a notable reduction in the logarithm of patents after the reform, although the exclusive dependence on Switzerland as a synthetic control limits the generalization of this result. The explanatory variables, such as GDP, population, net export balance, and public spending, present significant coefficients consistent with economic expectations, indicating their influence on patenting activity. Notably, public spending has a significant positive impact, highlighting its potential role in supporting innovation. Regarding the growth rate of the depreciation of IP assets, the DiD analyses do not indicate a significant post-reform effect. However, the SCM suggests a moderate increase in this rate after the reform. This observation may reflect a revaluation of the value of IP assets or a change in the accounting practices of Irish companies in response to the tax reform. Juxtaposing the results on patents and on the depreciation of IP assets allows us to hypothesize a link between the decrease in patenting activity and an increase in the depreciation of IP assets. It is possible that companies have modified their innovation strategy, favoring the internal exploitation of existing IP rather than filing new patents, thus leading to an increase in the depreciation of already held IP

assets. The different methods used in our study, although subject to limitations, offer a triangulation of results that reinforces the robustness of the observations. Despite the violation of the parallel trends assumption, the general trends observed across the DiD, Dynamic DiD, and SCM models are consistent, suggesting a real impact of the reform on innovation activity in Ireland.

In conclusion, although we cannot assert with certainty a causal relationship due to methodological constraints, the results obtained highlight significant changes associated with the Irish tax reform. These changes deserve particular attention and suggest that the reform has potentially influenced the innovation behaviors and management of IP assets of companies in Ireland. Further research, possibly with additional data or alternative methods, would be necessary to confirm these observations and deepen the understanding of the underlying mechanisms.

5.4.1 Summary of results and key findings

Method	Main Result <	Identified Limitation
Static DiD	Significant decrease in patents (-0.2766, $p < 0.01$). No effect for IP assets.	Violation of parallel trends
Dynamic DiD	Significant delayed effect on patents after 2 years (-0.2163). No effect on IP assets depreciation.	Absence of immediate effect observed. Violations of pre-treatment parallel trends.
SCM	Strong dependence on Switzerland (weight = 1.00) for patents. Strong dependence on Ireland (weight = 1.00) for the IP assets depreciation.	High RMSPE, cautious conclusions but justifies the choice of control for the static DiD

6 Conclusion

This study explored the impact of the introduction of the KDB in Ireland on innovation activity measured by the number of patents filed, as well as on the management of IP assets, evaluated by the growth rate of the depreciation of these assets. By employing robust econometric methods, notably static and dynamic DiD models, as well as the SCM, we sought to identify the effects of this tax reform on key innovation indicators.

The results obtained show a significant decrease in the number of patents filed after the introduction of the KDB, calling into question its effectiveness in stimulating innovation. These observations invite us to corroborate the criticisms of Griffith et al. (2014)¹¹, who note that patent boxes may incentivize tax relocation rather than the creation of genuine innovation. However, the divergent results of Bloom et al. (2002)⁸ on R&D credits highlight the importance of appropriate design of these mechanisms. In this sense, our conclusions call for a thorough examination of the eligibility criteria and monitoring mechanisms associated with the KDB. Furthermore, the increase in the depreciation rate of IP assets could reflect a corporate accounting strategy to maximize tax advantages. These findings raise crucial questions about the use of patent boxes as a public policy tool and necessitate additional research, particularly to test these hypotheses on sectoral or international samples.

The complexity of the underlying mechanisms of these observations underscores the limitations of tax policies as tools to stimulate innovation. Tax incentives, although important, cannot alone guarantee an increase in patenting activity or innovation. Other factors, such as the regulatory environment, availability of skills, research infrastructure, and collaborations between the public and private sectors, play a determining role in a country's innovative dynamism.

It is also important to acknowledge the methodological limitations of our study. The violation of the parallel trends assumption in the DiD models and the dependence on a single country in the SCM limit the robustness of the conclusions. Additionally, the restricted availability of post-reform data and the absence of data for certain comparable countries limit the generalization of the results.

6.1 Future research perspectives

To deepen the understanding of the impact of the KDB on innovation in Ireland, future research could explore several avenues. First, the use of microeconomic data at the firm level would allow for a more detailed analysis of innovation and patenting behaviors. Second, expanding the sample to include other countries or seeking alternative methods to overcome the limitations of parallel trends could strengthen the validity of the conclusions. Finally, exploring the longer-term effects of the reform, including data beyond 2018, could reveal delayed dynamics or strategic adjustments by companies that were not captured in the period studied.

In sum, this study highlights the need for a holistic approach to stimulate innovation, combining effective tax policies with investments in infrastructure, education, and cross-sector collaboration. Policymakers must consider these multiple dimensions to create an ecosystem conducive to sustainable innovation and economic growth.

7 Monte Carlo simulation : Analysis of the properties of the Synthetic Control Method

In this section, we conduct a Monte Carlo simulation to deeply analyze the statistical properties of the synthetic control method (SCM) presented earlier (see section^[4.4.1]). Building upon the framework of our study on patents, we create an artificial synthetic control by generating simulated data. The primary objective is to examine the distribution of key SCM statistics, notably the estimation of the treatment effect $\hat{\tau}$, the *Root Mean Square Prediction Error* (RMSPE), and the allocation of weights ω_k assigned to each control unit in the construction of the synthetic control.

This simulation pursues several fundamental objectives:

- **Methodological validation:** Evaluate the robustness and reliability of the SCM under controlled conditions by testing its ability to reproduce the dynamics of the treated country when the control group countries come from similar or identical distributions.
- **Exploration of Model Limitations:** Identify situations where the SCM may fail to produce reliable estimates, notably in the presence of excessive homogeneity among units, which can hinder the identification of causal effects.
- **Theoretical analysis:** Empirically demonstrate, through multiple iterations, that the estimation of the treatment effect $\hat{\tau}$ converges to zero when the characteristics of the control countries are indistinguishable from those of the treated country, thus confirming the absence of a causal effect under these conditions.
- **Study of weight allocation:** Analyze the distribution and assignment of weights ω_k to the control countries to understand how the SCM constructs the synthetic control in the absence of significant heterogeneity and how the weights influence the overall performance of the model.

Additionally, this simulation allows us to evaluate the impact of model specifications on its performance, particularly in terms of RMSPE, and to examine the behavior of the weights assigned to the control units. By analyzing the allocation of weights, we seek to understand the underlying mechanisms by which the SCM assigns importance to different units in constructing the synthetic control, especially when the control units are generated from the same distribution as the treated unit.

The results obtained contribute to the literature by validating the statistical foundations of the SCM and highlighting the conditions for optimal effectiveness of this methodology. By providing a detailed analysis of the treatment effect estimates, RMSPE, and weight allocation, we offer a comprehensive perspective on the performance and limitations of the SCM in a controlled simulation environment.

7.1 Presentation of the Monte Carlo approach

The simulation is based on random draws from a normal distribution, parameterized using the empirical characteristics of the real data for Ireland (mean = 5.899679 and standard deviation = 0.382714) (Table^[19]), considering the same time interval as the available data (1996-2018). We generated data from the treated country (Ireland) and a control group composed of 100 weighted fictitious countries. For each iteration, a random seed is set, ensuring the reproducibility of the experiments.

	patents	gdp	pop	NX	public_spending
mean	5.899679	0.021203	8.357304	0.078396	0.056326
std	0.382714	2.313380	0.098324	0.133779	0.050162

Table 19: Descriptive statistics of interest for Ireland (2000-2019), extracted from Table ^[1] (units description in section^[3.1])

In each simulation:

- Using a mean and standard deviation as indicated in Table ^[19], we randomly draw the dependent variable **Patents_{it}** and the explanatory variables **X_{k,it}** following three models:

1. Simulation of initial Data:

- Dependent variable: $\text{Patents}_{it} \sim \mathcal{N}(\mu_{[\text{Patents, IRL}]}, \sigma_{[\text{Patents, IRL}]}^2)$
- Explanatory variables: $X_{k,it} \sim \mathcal{N}(\mu_{[X_k, \text{IRL}]}, \sigma_{[X_k, \text{IRL}]}^2)$

2. Simulation of heteroskedastic data:

- Dependent variable: $\text{Patents}_{it} \sim \mathcal{N}(\mu_{[\text{Patents, IRL}]}, \sigma_{[\text{Patents, IRL}]}^2)$
- Explanatory variables:
$$X_{k,it} \sim \mathcal{N}(\mu_{[X_k, \text{IRL}]} + \epsilon_k, \sigma_{[X_k, \text{IRL}]}^2 + \delta_k) \quad \left\{ \begin{array}{l} \epsilon_k \sim \mathcal{N}(0, \sigma_{\epsilon_k}^2), \\ \sigma_{\epsilon_k}^2 \sim \mathcal{U}(0.0025, 0.0225), \\ \delta_k = (\mu_{[X_k, \text{IRL}]} \times \sigma_{[X_k, \text{IRL}]})^2 - \sigma_{[X_k, \text{IRL}]}^2 \end{array} \right.$$

3. Simulation of correlated data:

- All variables are simulated jointly according to a multivariate normal distribution:

$$(\text{Patents}_{it}, X_{1,it}, \dots, X_{k,it}) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad \left\{ \begin{array}{l} \boldsymbol{\mu} = (\mu_{[\text{Patents, IRL}]}, \mu_{[X_1, \text{IRL}]}, \dots, \mu_{[X_k, \text{IRL}]}) \\ \boldsymbol{\Sigma} = \text{Covariance matrix with strong correlations} \end{array} \right.$$

$\boldsymbol{\Sigma}$ is the covariance matrix constructed with strong random correlations between -0.9 and -0.7 or between 0.7 and 0.9 , adjusted to be positive semi-definite.

- We perform the SCM optimization: $\min_{\omega} (\mathbf{X}_{\text{IRL}} - \sum_{k=1}^{100} \omega_k \mathbf{X}_k)' \mathbf{V} (\mathbf{X}_{\text{IRL}} - \sum_{k=1}^{100} \omega_k \mathbf{X}_k) \quad \text{s.t.} \quad \left\{ \sum_{k=1}^{100} \omega_k = 1, \omega_k \geq 0 \quad \forall k \right.$ to obtain $\hat{\mathbf{y}}_{\text{IRL},t}^{\text{Synth}} = \sum_{k=1}^{100} \omega_k \cdot \mathbf{y}_{k,t}$
- The SCM model is used to evaluate $\hat{\tau} = \mathbf{y}_{\text{IRL},2016} - \hat{\mathbf{y}}_{\text{IRL},2016}^{\text{Synth}}$ and the RMSPE = $\sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{y}_{\text{synthetic},t} - y_{\text{treated},t})^2}$.

The results of the simulations are then aggregated to construct and analyze the distributions of $\hat{\tau}$ and the RMSPE through histograms and descriptive statistics.

7.2 Results

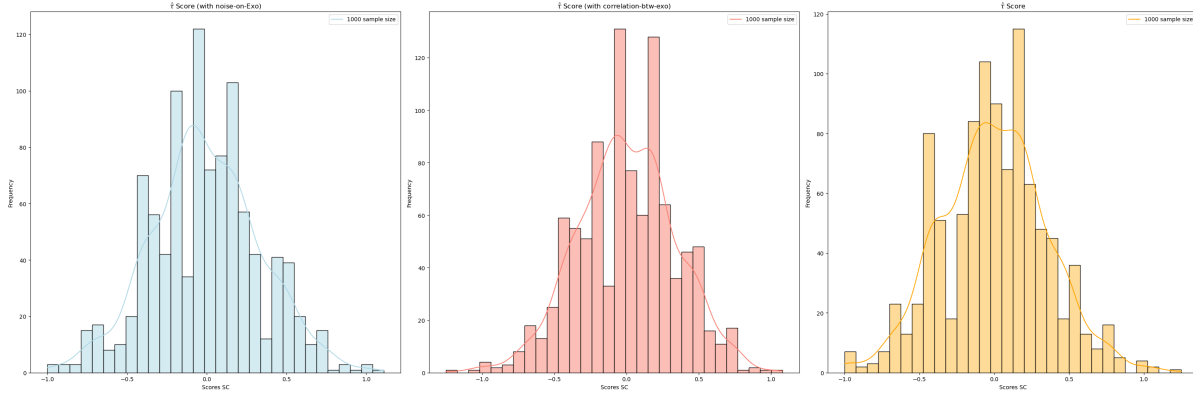


Figure 7: Histogram of parameters $\hat{\tau}$ (blue: normal draw with noise, red: draw with correlated exogenous variables, orange: normal draw)

	count	mean	std	min	25%	50%	75%	max	skew	kurt
noise	1000	-0.008280	0.340096	-1.000000	-0.212500	-0.025000	0.182500	1.110000	2.828412	8.999735
correlation	1000	-0.000020	0.339414	-1.250000	-0.222500	0.000000	0.210000	1.080000	2.828408	8.999716
normal	1000	-0.003150	0.353006	-1.000000	-0.240000	-0.000000	0.230000	1.250000	2.828409	8.999730

Table 20: Descriptive statistics for the SCM simulation

The results in Table^[20] demonstrate that the estimates of $\hat{\tau}$ for the three types of simulations—heteroskedasticity, correlated data, and normal draw—are on average very close to zero. This illustrates the convergence of the synthetic control method (SCM) towards the absence of a causal effect and confirms the absence of systematic bias. This convergence is also supported by the Central Limit Theorem, which predicts that the mean of the estimates, in large numbers, will be normally distributed around the true parameter value. The consistency of the standard deviation, around 0.34 to 0.35, indicates comparable variability of the estimates regardless of the structure of the simulated data, demonstrating the robustness of the SCM. The measures of skewness and kurtosis, with skewness around 2.828 and kurtosis of 9 for all simulations, reveal a slightly right-skewed and leptokurtic distribution, indicating heavier tails than the normal distribution. These distributions, along with the minimum and maximum values of $\hat{\tau}$, which show some dispersion but with quartiles centered around zero, reinforce the idea that the majority of estimates concentrate near the absence of effect. These observations confirm the reliability and robustness of the SCM in producing unbiased estimates of the treatment effect in a simulation framework, thus validating its use for causal analysis.

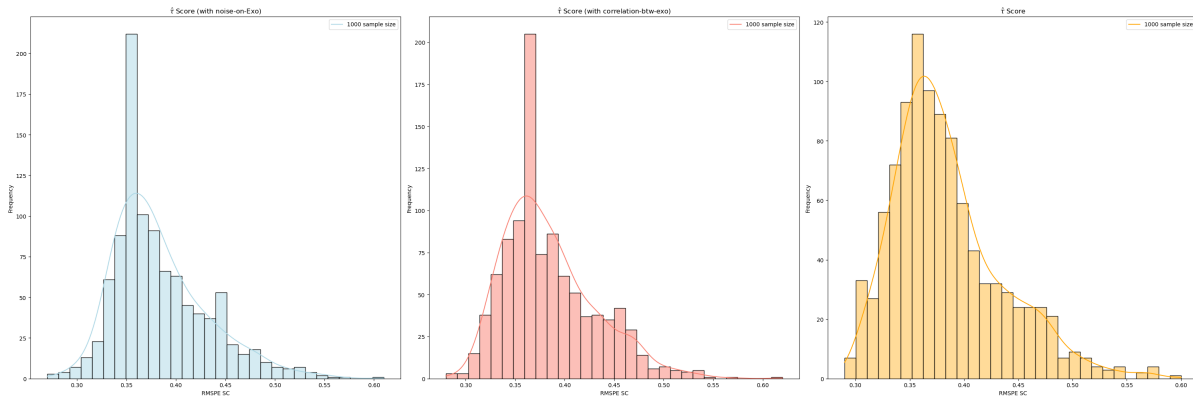


Figure 8: Histogram of *RMSPE* parameters (blue: normal draw with noise, red: draw with correlated exogenous variables, orange: normal draw)

	count	mean	std	min	25%	50%	75%	max	skew	kurt
noise	1000	0.383880	0.048167	0.270000	0.350000	0.370000	0.410000	0.610000	2.828426	8.999839
correlation	1000	0.384170	0.047097	0.280000	0.350000	0.370000	0.410000	0.620000	2.828426	8.999839
normal	1000	0.385250	0.049954	0.290000	0.350000	0.370000	0.410000	0.600000	2.828426	8.999840

Table 21: Descriptive statistics for the RMSPE simulation

Table ^[21] presents the descriptive statistics of the *Root Mean Square Prediction Error* (RMSPE) for the three types of simulations: 'noise' (heteroskedastic data), 'correlation' (correlated data), and 'normal' (simple normal draws). The RMSPE means are very close for the three configurations (respectively 0.3839, 0.3842, and 0.3853), indicating consistent predictive performance of the SCM regardless of the structure of the simulated data and demonstrating the robustness of the model. The standard deviation (std), about 0.048 in each simulation, reflects comparable variability of the RMSPE, while the minimum and maximum values, between 0.27 and 0.62, show moderate dispersion. Furthermore, the quartiles (25%, 50%, 75%) are constant across the different simulations, confirming the stability of the predictive performance of the SCM, regardless of the structure of the simulated data. The skewness and kurtosis are high (skewness ≈ 2.828 , kurtosis ≈ 9) for all simulations. This strong positive skewness can be attributed to the fact that the RMSPE, being an always positive measure of the mean squared error, follows a right-skewed distribution with heavy tails. This is consistent with the statistical properties of variances, which, being squares of deviations, are always positive and can exhibit significant variability. These results confirm the convergence of the RMSPE toward a stable value as the number of iterations increases, in accordance with the Central Limit Theorem. The constancy of the RMSPE statistics across the different simulations underscores the robustness of the SCM in the face of variations in the data structure, reinforcing the reliability of this model for estimating treatment effects under various conditions.

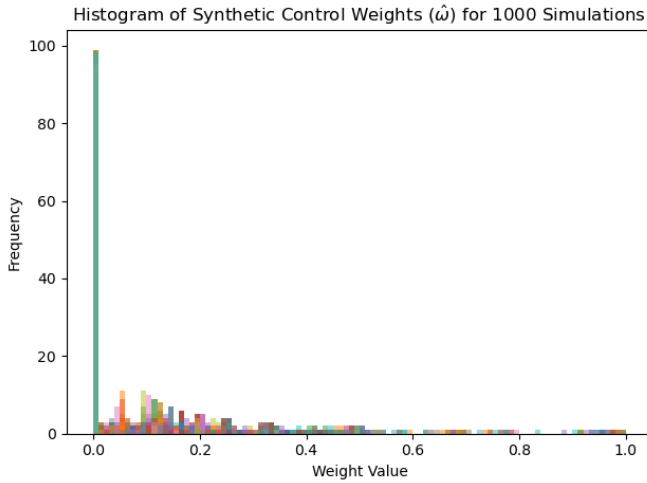


Figure 9: SC Weights allocation by frequency—Normal 1000 draws

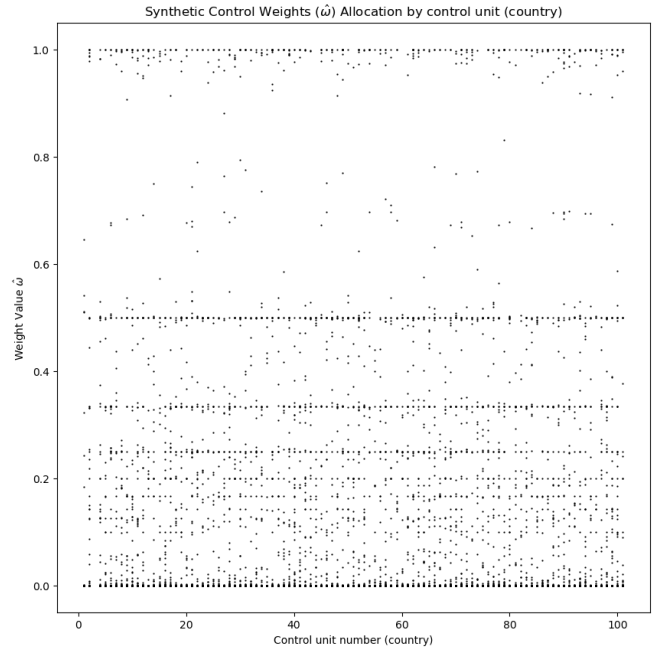


Figure 10: SC Weights value allocation by control (country)—Normal 1000 draws

	weights_mean	std	skewness	kurtosis	median
mean of all	0.061419	1.630565	2.809768	8.911182	0.831493

Table 22: Descriptive statistics for the allocation of weights $\hat{\omega}$ for 100 draws

Table^[22] presents the descriptive statistics of the allocation of weights $\hat{\omega}$ from 1000 normal simulations of the SCM, while

Figure^[10] illustrates the distribution of these weights by control unit. The mean of the weights is low (weights_mean = 0.0614), while the median is significantly higher (median = 0.8315), indicating a highly skewed distribution, as confirmed by the high skewness coefficient (skewness = 2.8098) and high kurtosis (kurtosis = 8.9112).

These results reveal that the majority of weights $\hat{\omega}$ are equal to or close to zero, as shown by the significant concentration of weights around 0 in Figure^[9]. However, a few control units stand out with significantly higher weights, reaching up to 1, reflecting the parsimony of the SCM in selecting the most relevant units. The distinct vertical structure of the points in the graph indicates that certain groups of units receive similar weights, suggesting a strong similarity between these units and the treated unit. This characteristic distribution of the SCM highlights its ability to optimize the synthetic combination by primarily assigning high weights to controls capable of faithfully reproducing the pre-intervention trajectory of the treated unit, while assigning zero weights to less relevant units. The positive skewness and high kurtosis highlight the presence of extreme values, where a few very high weights increase the mean but remain rare compared to the majority of weights near zero.

Thus, the combination of analyses from Table^[22], Figure^[9], and Figure^[10] visually and statistically validates the parsimonious property of the SCM. These observations reinforce the interpretation that the SCM favors a weight allocation concentrated on a restricted subset of relevant units, thereby strengthening the validity of the causal estimates obtained by this method.

7.3 Conclusion

The Monte Carlo simulation conducted in this study allowed us to illustrate both the parameters and the fundamental assumptions of the synthetic control method (SCM). By generating simulated data according to different structures (normal, heteroskedastic, and correlated), we empirically demonstrated that the SCM produces estimates of the treatment effect $\hat{\tau}$ close to zero in the absence of a real causal effect. This convergence toward a centered normal distribution validates the fundamental assumption of the SCM that, without intervention, the differences between the treated unit and the synthetic control are random and not systematic.

The analysis of the *Root Mean Square Prediction Error* (RMSPE) showed stable predictive performance of the SCM, regardless of variations in the structure of the simulated data. This robustness confirms the reliability of the model in faithfully reproducing the pre-intervention trajectory of the treated unit, even when data are generated from identical distributions.

Moreover, the study of the weight allocation ω_k highlighted the parsimonious property of the SCM. The concentration of weights on a restricted subset of relevant control units underscores the model's ability to select the units most similar to the treated unit, thus optimizing the construction of the synthetic control.

In summary, this Monte Carlo simulation validates the theoretical foundations of the SCM by illustrating its performance in a controlled framework. It confirms that the SCM maintains unbiased estimates of $\hat{\tau}$ and stable predictive performance, even when the control units are generated from the same distribution as the treated unit. These results also highlight the importance of heterogeneity between the treated group and the control group for the precise identification of causal effects, thus underscoring the necessary conditions for the optimal effectiveness of the SCM. This heterogeneity must be present up to the limit of validating the parallel trends assumption. Indeed, excessive heterogeneity could violate this fundamental assumption of the SCM, which assumes that, without treatment, the trajectories of treated and untreated units would have evolved similarly. It is therefore crucial to find a balance between sufficient heterogeneity to identify significant causal effects and sufficient similarity to respect the parallel trends assumption, thus ensuring the validity of the estimates obtained by the SCM.

8 Appendix

8.1 Visual trends for control selection^[3]

8.1.1 Patents

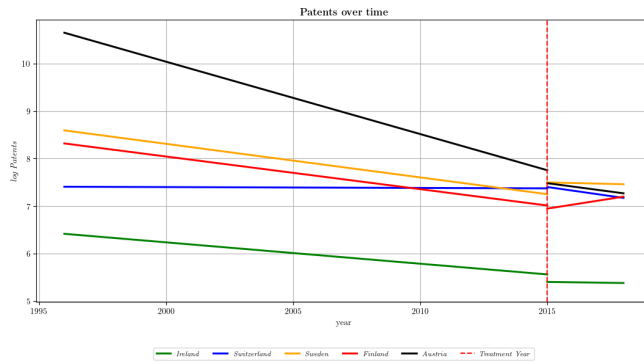


Figure 11: Simple discontinuity regression

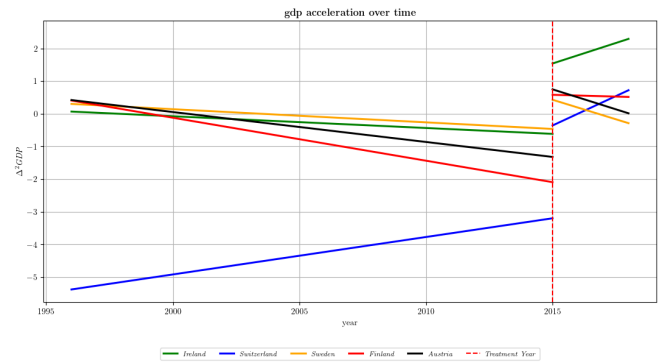


Figure 12: Simple discontinuity regression

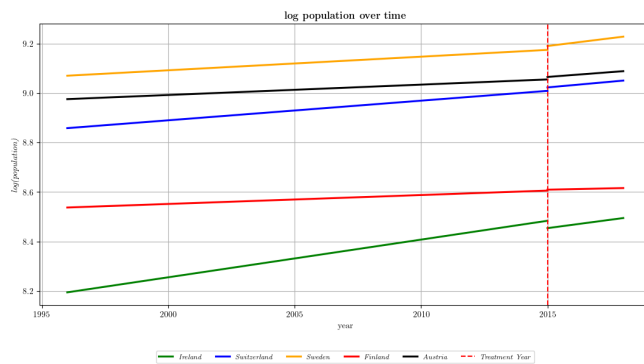


Figure 13: Simple discontinuity regression

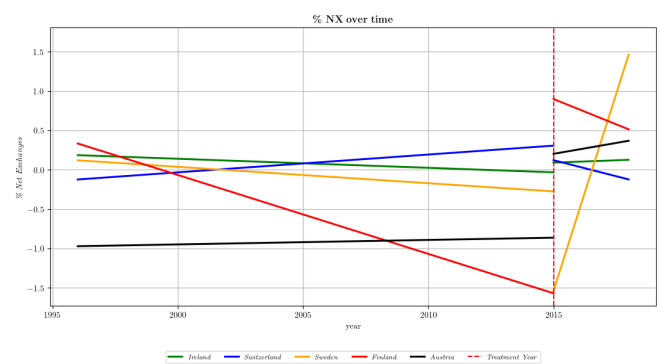


Figure 14: Simple discontinuity regression

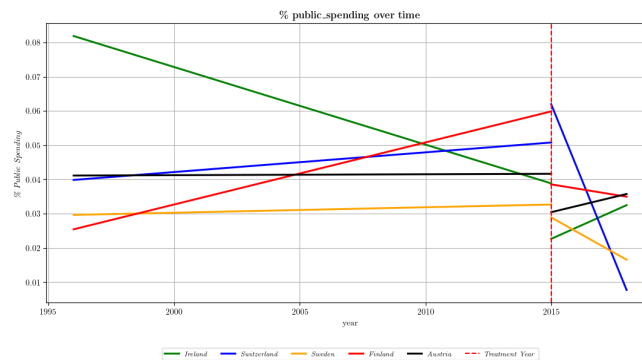


Figure 15: Simple discontinuity regression

8.1.2 Depreciation of IP assets

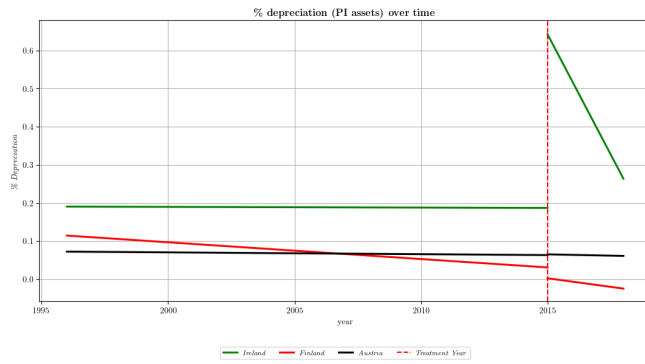


Figure 16: Simple discontinuity regression



Figure 17: Simple discontinuity regression



Figure 18: Simple discontinuity regression

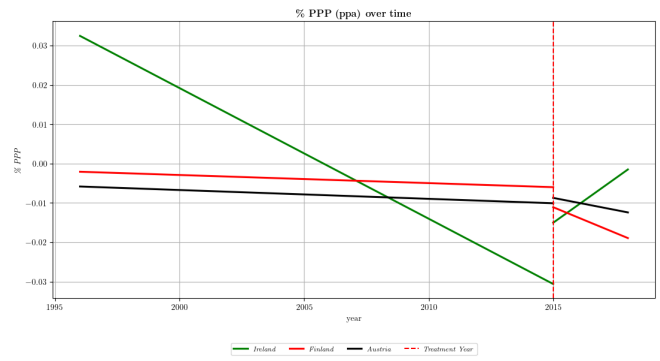


Figure 19: Simple discontinuity regression

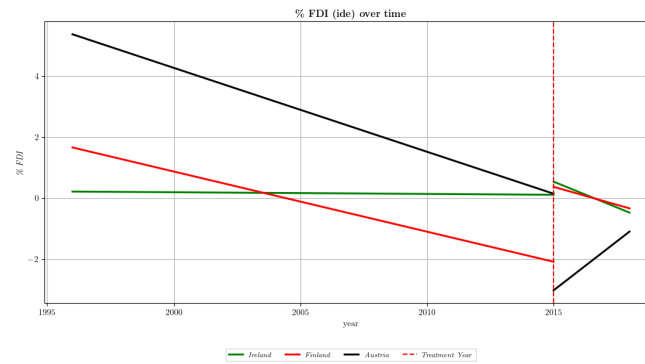


Figure 20: Simple discontinuity regression

8.2 Parallel trend test summary

8.2.1 Patents ^[5.1.1]

Dep. Variable:	brevets	R-squared:	0.978
Model:	OLS	Adj. R-squared:	0.947
Method:	Least Squares	F-statistic:	31.93
Date:	Mon, 25 Nov 2024	Prob (F-statistic):	7.23e-11
Time:	16:39:20	Log-Likelihood:	33.161
No. Observations:	46	AIC:	-12.32
Df Residuals:	19	BIC:	37.05
Df Model:	26		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	7.3979	0.142	52.276	0.000	7.102	7.694
C(year)[T.1997]	0.0624	0.183	0.341	0.737	-0.321	0.446
C(year)[T.1998]	0.0402	0.183	0.219	0.829	-0.344	0.424
C(year)[T.1999]	-0.0205	0.184	-0.111	0.912	-0.405	0.364
C(year)[T.2000]	-0.0225	0.184	-0.122	0.904	-0.408	0.363
C(year)[T.2001]	0.0408	0.185	0.220	0.828	-0.347	0.428
C(year)[T.2002]	-0.0279	0.186	-0.150	0.882	-0.417	0.361
C(year)[T.2003]	-0.1005	0.187	-0.538	0.597	-0.492	0.291
C(year)[T.2004]	-0.1551	0.188	-0.824	0.420	-0.549	0.239
C(year)[T.2005]	-0.0435	0.189	-0.229	0.821	-0.440	0.353
C(year)[T.2006]	0.0566	0.191	0.296	0.770	-0.343	0.456
C(year)[T.2007]	0.0142	0.193	0.074	0.942	-0.389	0.417
C(year)[T.2008]	0.0867	0.194	0.446	0.660	-0.320	0.493
C(year)[T.2009]	0.1183	0.196	0.603	0.554	-0.292	0.529
C(year)[T.2010]	0.1343	0.198	0.678	0.506	-0.281	0.549
C(year)[T.2011]	-0.0346	0.200	-0.173	0.865	-0.454	0.385
C(year)[T.2012]	-0.0764	0.203	-0.377	0.710	-0.501	0.348
C(year)[T.2013]	-0.1474	0.205	-0.719	0.481	-0.577	0.282
C(year)[T.2014]	-0.0922	0.207	-0.444	0.662	-0.526	0.342
C(year)[T.2015]	0.2201	0.146	1.510	0.148	-0.085	0.525
C(year)[T.2016]	-0.1077	0.121	-0.893	0.383	-0.360	0.145
C(year)[T.2017]	-0.1949	0.121	-1.616	0.123	-0.447	0.058
C(year)[T.2018]	-0.0060	0.146	-0.041	0.968	-0.311	0.299
C(country)[T.IRL]	43.0783	10.873	3.962	0.001	20.321	65.836
treated	43.0783	10.873	3.962	0.001	20.321	65.836
post	-0.0884	0.135	-0.656	0.520	-0.371	0.194
treated:post	-227.0819	234.524	-0.968	0.345	-717.947	263.783
treated_pre_time	-0.0437	0.011	-4.026	0.001	-0.066	-0.021
treated_post_time	0.0689	0.116	0.595	0.559	-0.173	0.311

Omnibus:	1.334	Durbin-Watson:	1.004
Prob(Omnibus):	0.513	Jarque-Bera (JB):	1.007
Skew:	-0.000	Prob(JB):	0.605
Kurtosis:	2.275	Cond. No.	2.88e+19

Table 23: Patents: Parallel trends test results summary

8.2.2 Depreciation of IP assets^[5.1.2]

Dep. Variable:	depreciation	R-squared:	0.985
Model:	OLS	Adj. R-squared:	0.964
Method:	Least Squares	F-statistic:	47.25
Date:	Mon, 25 Nov 2024	Prob (F-statistic):	2.02e-12
Time:	17:11:54	Log-Likelihood:	125.38
No. Observations:	46	AIC:	-196.8
Df Residuals:	19	BIC:	-147.4
Df Model:	26		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.0447	0.019	2.343	0.030	0.005	0.085
C(year)[T.1997]	0.0211	0.025	0.854	0.404	-0.031	0.073
C(year)[T.1998]	0.0331	0.025	1.340	0.196	-0.019	0.085
C(year)[T.1999]	0.0474	0.025	1.914	0.071	-0.004	0.099
C(year)[T.2000]	0.0841	0.025	3.387	0.003	0.032	0.136
C(year)[T.2001]	0.0791	0.025	3.173	0.005	0.027	0.131
C(year)[T.2002]	0.0502	0.025	2.005	0.059	-0.002	0.103
C(year)[T.2003]	0.0661	0.025	2.623	0.017	0.013	0.119
C(year)[T.2004]	0.0798	0.025	3.148	0.005	0.027	0.133
C(year)[T.2005]	0.0713	0.026	2.792	0.012	0.018	0.125
C(year)[T.2006]	0.0471	0.026	1.832	0.083	-0.007	0.101
C(year)[T.2007]	0.0622	0.026	2.398	0.027	0.008	0.116
C(year)[T.2008]	0.0684	0.026	2.615	0.017	0.014	0.123
C(year)[T.2009]	0.0058	0.026	0.218	0.830	-0.050	0.061
C(year)[T.2010]	-0.0179	0.027	-0.669	0.512	-0.074	0.038
C(year)[T.2011]	-0.0396	0.027	-1.468	0.158	-0.096	0.017
C(year)[T.2012]	0.0155	0.027	0.567	0.577	-0.042	0.073
C(year)[T.2013]	-0.0234	0.028	-0.849	0.407	-0.081	0.034
C(year)[T.2014]	-0.0449	0.028	-1.606	0.125	-0.103	0.014
C(year)[T.2015]	0.0100	0.020	0.512	0.615	-0.031	0.051
C(year)[T.2016]	-0.0150	0.016	-0.924	0.367	-0.049	0.019
C(year)[T.2017]	-0.0220	0.016	-1.352	0.192	-0.056	0.012
C(year)[T.2018]	-0.0177	0.020	-0.899	0.380	-0.059	0.023
C(country)[T.IRL]	4.7010	1.464	3.210	0.005	1.636	7.766
treated	4.7010	1.464	3.210	0.005	1.636	7.766
post	-0.0446	0.018	-2.455	0.024	-0.083	-0.007
treated:post	227.6152	31.588	7.206	0.000	161.500	293.731
treated_pre_time	-0.0046	0.001	-3.180	0.005	-0.008	-0.002
treated_post_time	-0.1173	0.016	-7.521	0.000	-0.150	-0.085

Omnibus:	0.065	Durbin-Watson:	1.978
Prob(Omnibus):	0.968	Jarque-Bera (JB):	0.022
Skew:	-0.000	Prob(JB):	0.989
Kurtosis:	2.892	Cond. No.	3.09e+19

Table 24: Depreciation IP Assets: Parallel trends test results summary

8.3 Dynamic DiD (event study) summary

8.3.1 Patents ^[5.2.2]

Dependent Variable:	brevets
Model:	(1)
<i>Variables</i>	
event_time = -19	-0.4217* (0.0644)
event_time = -18	-0.4782 (0.3663)
event_time = -17	-0.3896 (0.3684)
event_time = -16	-0.3349 (0.3586)
event_time = -15	-0.4846 (0.0972)
event_time = -14	-0.7462 (0.1398)
event_time = -13	-0.8450* (0.1162)
event_time = -12	-0.5585 (0.2228)
event_time = -11	-0.7258 (0.2577)
event_time = -10	-0.3487 (0.4188)
event_time = -9	-0.4290 (0.4954)
event_time = -8	-0.4750 (0.4473)
event_time = -7	-0.3057 (0.1947)
event_time = -6	-0.4272 (0.3565)
event_time = -5	0.0224 (0.1119)
event_time = -4	-0.1325 (0.2724)
event_time = -3	-0.1864 (0.3478)
event_time = -2	-0.1601 (0.2611)
event_time = 0	-0.0654 (0.2882)
event_time = 1	-0.2866 (0.1298)
event_time = 2	-0.2163 (0.0397)
event_time = 3	0.0603 (0.1720)
gdp	-0.0026*** (6.8×10^{-17})
pop	-3.912*** (7.32×10^{-14})
NX	-0.0303*** (9.51×10^{-16})
public_spending	5.724*** (5.07×10^{-14})
<i>Fixed-effects</i>	
country	Yes
<i>Fit statistics</i>	
Observations	46
R ²	0.98268
Within R ²	0.86140
<i>Clustered (country) standard-errors in parentheses</i>	
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>	

Table 25: Dynamic DiD event study summary

8.3.2 Depreciation of IP assets^[5.2.4]

Dependent Variable: Model:	depreciation (1)
<i>Variables</i>	
event_time = -19	0.0873 (0.0723)
event_time = -18	0.1165* (0.0108)
event_time = -17	0.1193 (0.0278)
event_time = -16	0.1132 (0.1311)
event_time = -15	0.1413 (0.0286)
event_time = -14	0.1459** (0.0111)
event_time = -13	0.2257 (0.0656)
event_time = -12	0.1852 (0.0857)
event_time = -11	0.1061 (0.0488)
event_time = -10	0.0785 (0.0376)
event_time = -9	0.0610 (0.0785)
event_time = -8	0.1494 (0.1281)
event_time = -7	0.1733 (0.1874)
event_time = -6	0.2795 (0.1465)
event_time = -5	0.0427 (0.0092)
event_time = -4	0.0204 (0.0128)
event_time = -3	0.1076 (0.0824)
event_time = -2	0.0398 (0.0362)
event_time = 0	0.3267 (0.3877)
event_time = 1	0.1509 (0.3277)
event_time = 2	0.1409 (0.1880)
event_time = 3	0.0385 (0.2294)
salary	0.9891*** (2.87×10^{-14})
cho	-0.4850*** (9.07×10^{-16})
ppa	-1.722*** (1.46×10^{-14})
IDE	-0.0168*** (3.9×10^{-16})
<i>Fixed-effects</i>	
country	Yes
<i>Fit statistics</i>	
Observations	46
R ²	0.71628
Within R ²	0.56062
<i>Clustered (country) standard-errors in parentheses</i>	
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>	

Table 26: Dynamic DiD event study summary

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